1. Crackme 1
   1.1. Download file
   1.2.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ ./ crackme1
zsh: permission denied: ./

┌──(kali㉿kali)-[~/Downloads]
└─$ ls -l
total 498212
-rw-rw-r--   1 kali kali      7192 Apr  8 11:07 crackme1
drwxr-xr-x 10 kali kali      4096 Apr  7 22:33 ghidra_12.0.4_PUBLIC
-rw-rw-r--   1 kali kali 509952513 Apr  7 21:12 ghidra_12.0.4_PUBLIC_20260303.zip
-rw-rw-r--   1 kali kali    176913 Apr  7 22:46 test.txt
-rw-rw-r--   1 kali kali      6099 Mar 24 20:49 us-east-1-maki.andrich-premium.ovpn
-rw-rw-r--   1 kali kali      6099 Apr  1 21:10 us-west-2-maki.andrich-premium.ovpn

┌──(kali㉿kali)-[~/Downloads]
└─$ chmod +x crackme1

┌──(kali㉿kali)-[~/Downloads]
└─$ ./ crackme1
zsh: permission denied: ./
```

      1.2.1. Still doesnt execute lol bc space
   1.3.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ ./crackme1
flag{not_that_kind_of_elf}
```

      1.3.1. Here we go
   1.4. Flag is
      1.4.1. Flag{not_that_kind_of_elf}
2. Crackme 2
   2.1. Download the file and test it out

2.2.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ ls -l
total 498220
-rwxrwxr-x  1 kali kali      7192 Apr  8 11:07 crackme1
-rw-rw-r--  1 kali kali      5884 Apr  8 11:17 crackme2
drwxr-xr-x 10 kali kali      4096 Apr  7 22:33 ghidra_12.0.4_PUBLIC
-rw-rw-r--  1 kali kali 509952513 Apr  7 21:12 ghidra_12.0.4_PUBLIC_20260303.zip
-rw-rw-r--  1 kali kali    176913 Apr  7 22:46 test.txt
-rw-rw-r--  1 kali kali      6099 Mar 24 20:49 us-east-1-maki.andrich-premium.ovpn
-rw-rw-r--  1 kali kali      6099 Apr  1 21:10 us-west-2-maki.andrich-premium.ovpn

┌──(kali㉿kali)-[~/Downloads]
└─$ chmod +x crackme2

┌──(kali㉿kali)-[~/Downloads]
└─$ ls -l
total 498220
-rwxrwxr-x  1 kali kali      7192 Apr  8 11:07 crackme1
-rwxrwxr-x  1 kali kali      5884 Apr  8 11:17 crackme2
drwxr-xr-x 10 kali kali      4096 Apr  7 22:33 ghidra_12.0.4_PUBLIC
-rw-rw-r--  1 kali kali 509952513 Apr  7 21:12 ghidra_12.0.4_PUBLIC_20260303.zip
-rw-rw-r--  1 kali kali    176913 Apr  7 22:46 test.txt
-rw-rw-r--  1 kali kali      6099 Mar 24 20:49 us-east-1-maki.andrich-premium.ovpn
-rw-rw-r--  1 kali kali      6099 Apr  1 21:10 us-west-2-maki.andrich-premium.ovpn

┌──(kali㉿kali)-[~/Downloads]
└─$ ./crackme2
Usage: ./crackme2 password

┌──(kali㉿kali)-[~/Downloads]
└─$ ./crackme2 test
Access denied.

┌──(kali㉿kali)-[~/Downloads]
└─$
```

2.2.1. We need a password as the second parameter

2.2.2. Instead of brute force lets look at the exe to see if we can find what its checking against

2.3.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ strings crackme2 > test.txt
```

2.4. Lets investigate

2.5.

```
1 /lib/ld-linux.so.2
2 libc.so.6
3 _IO_stdin_used
4 puts
5 printf
6 memset
7 strcmp
```

2.5.1. String compare so lets see if there are any strings in here to try

2.6.

```
1 /lib/ld-linux.so.2
2 libc.so.6
3 _IO_stdin_used
4 puts
5 printf
6 memset
7 strcmp
8 __libc_start_main
9 /usr/local/lib:$ORIGIN
10 __gmon_start__
11 GLIBC_2.0
12 PTRh
13 j3jA
14 [^_]
15 UWVS
16 t$,U
17 [^_]
18 Usage: %s password
19 super_secret_password
20 Access denied.
21 Access granted.
22 ;*2$"(
23 GCC: (Ubuntu 5.4.0-6ubuntu1~16.04.9) 5.4.0 20160609
24 crtstuff.c
25 __JCR_LIST__
26 deregister_tm_clones
27 __do_global_dtors_aux
28 completed.7209
29 __do_global_dtors_aux_fini_array_entry
30 frame_dummy
31 __frame_dummy_init_array_entry
32 conditional1.c
33 giveFlag
34 __FRAME_END__
35 __JCR_END__
36 __init_array_end
37 _DYNAMIC
38 __init_array_start
39 __GNU_EH_FRAME_HDR
40 _GLOBAL_OFFSET_TABLE_
41 __libc_csu_fini
42 strcmp@@GLIBC_2.0
43 _ITM_deregisterTMCloneTable
```

2.6.1. Super_secret_password

2.6.2. Seems promising

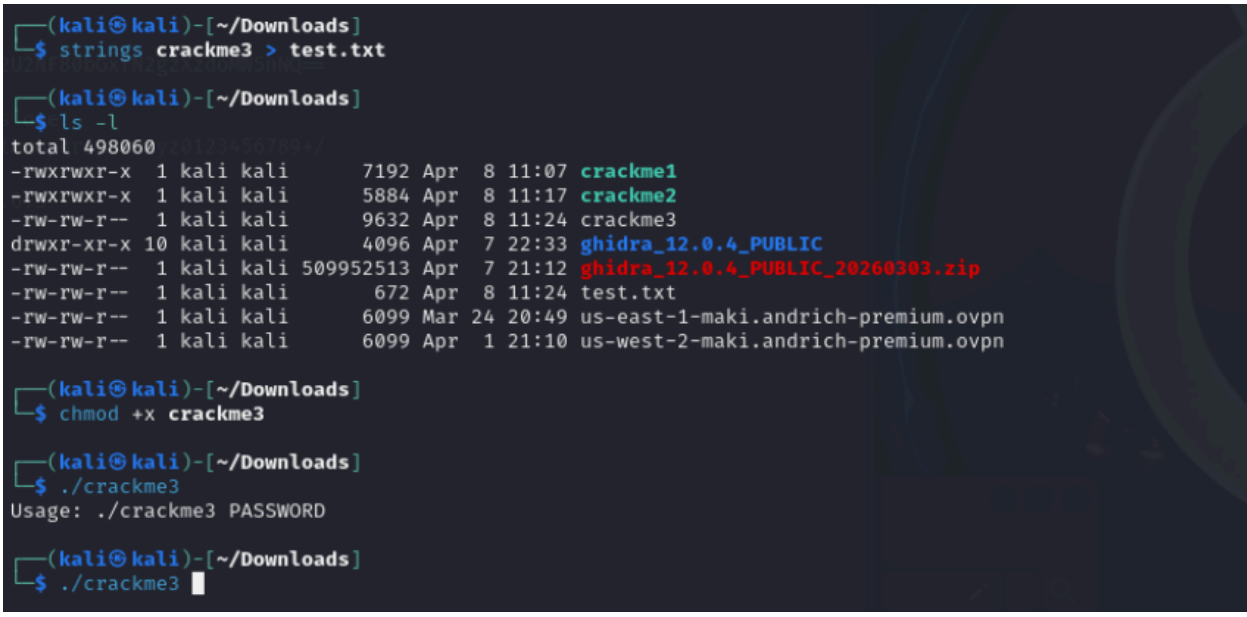
2.7.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ ./crackme2 super_secret_password
Access granted.
flag{if_i_submit_this_flag_then_i_will_get_points}
```

2.7.1. Flag{if_i_submit_this_flag_then_i_will_get_points}

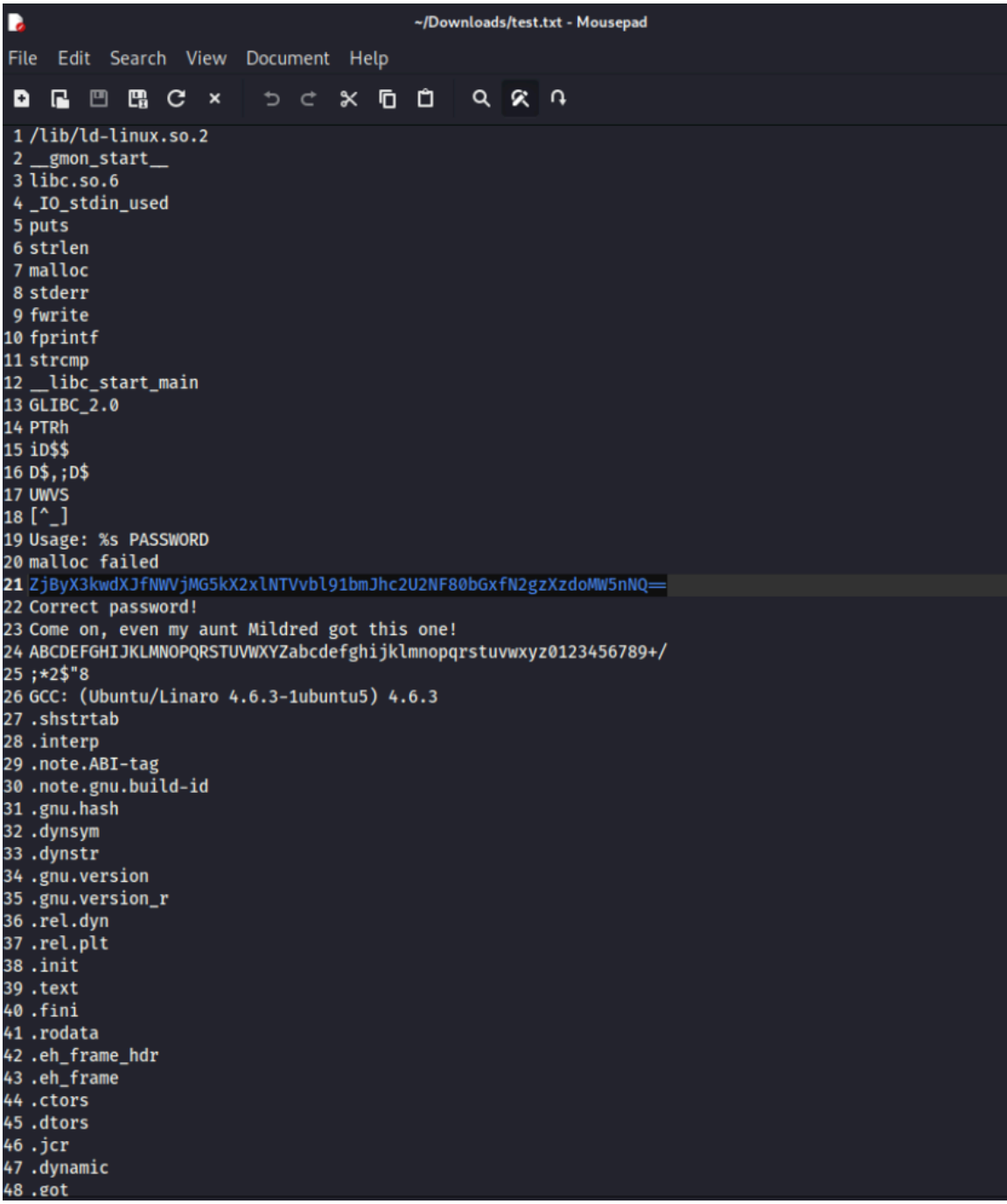
3. Crackme 3

3.1.



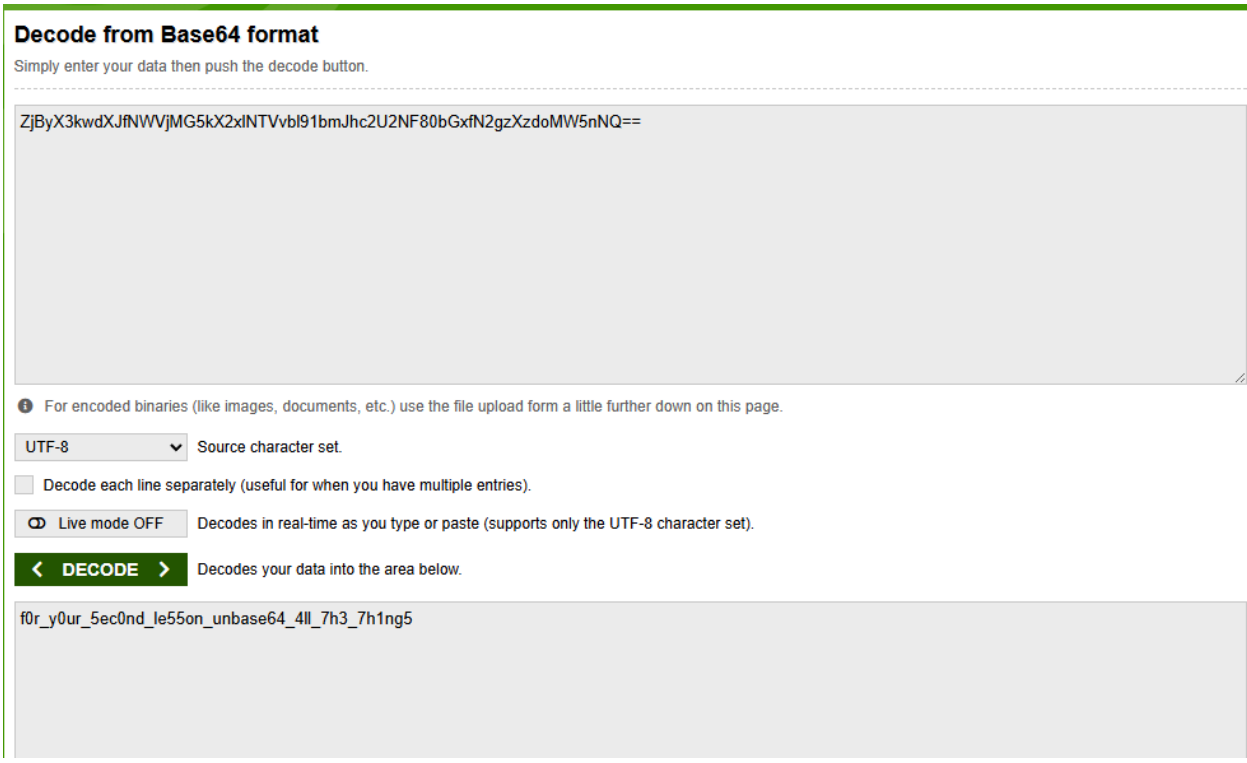
3.1.1. Same thing you need a password

3.2. Lets Look into the strings



3.2.1.

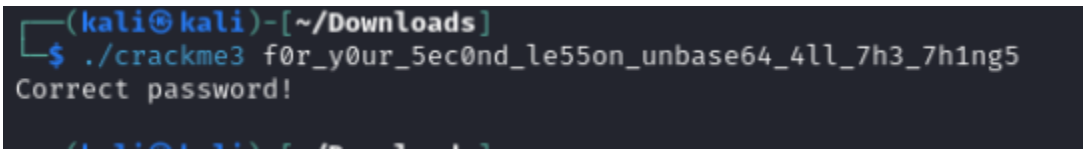
3.2.1.1. This looks incoded

3.3. 

3.3.1. Bang it was base64 and we get

3.3.1.1. F0r_y0ur_5ec0nd_le55on_unbase64_4ll_7h3_7h1ng5

3.4. Use that as the password

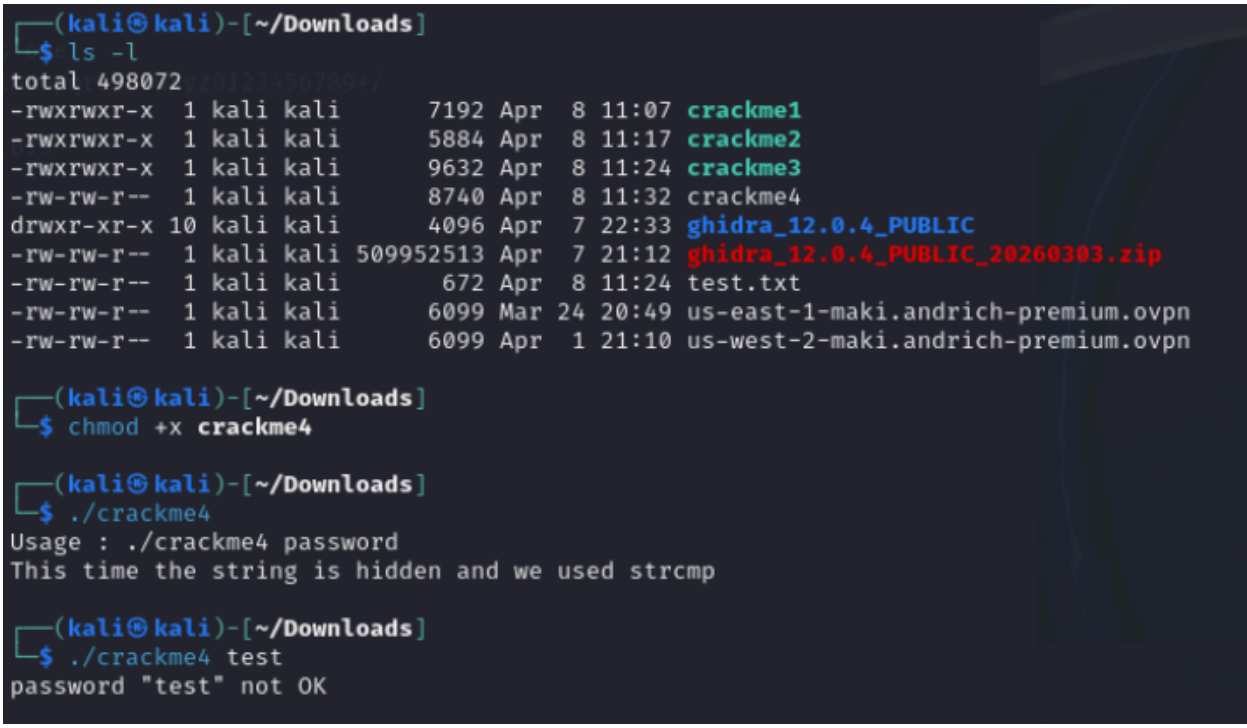
3.4.1. 

3.4.2. Correct password but no flag…

3.5. Wait that was our flag lol

4. Crackme 4

4.1.



4.2. Okay so the string is hidden and they used strcmp

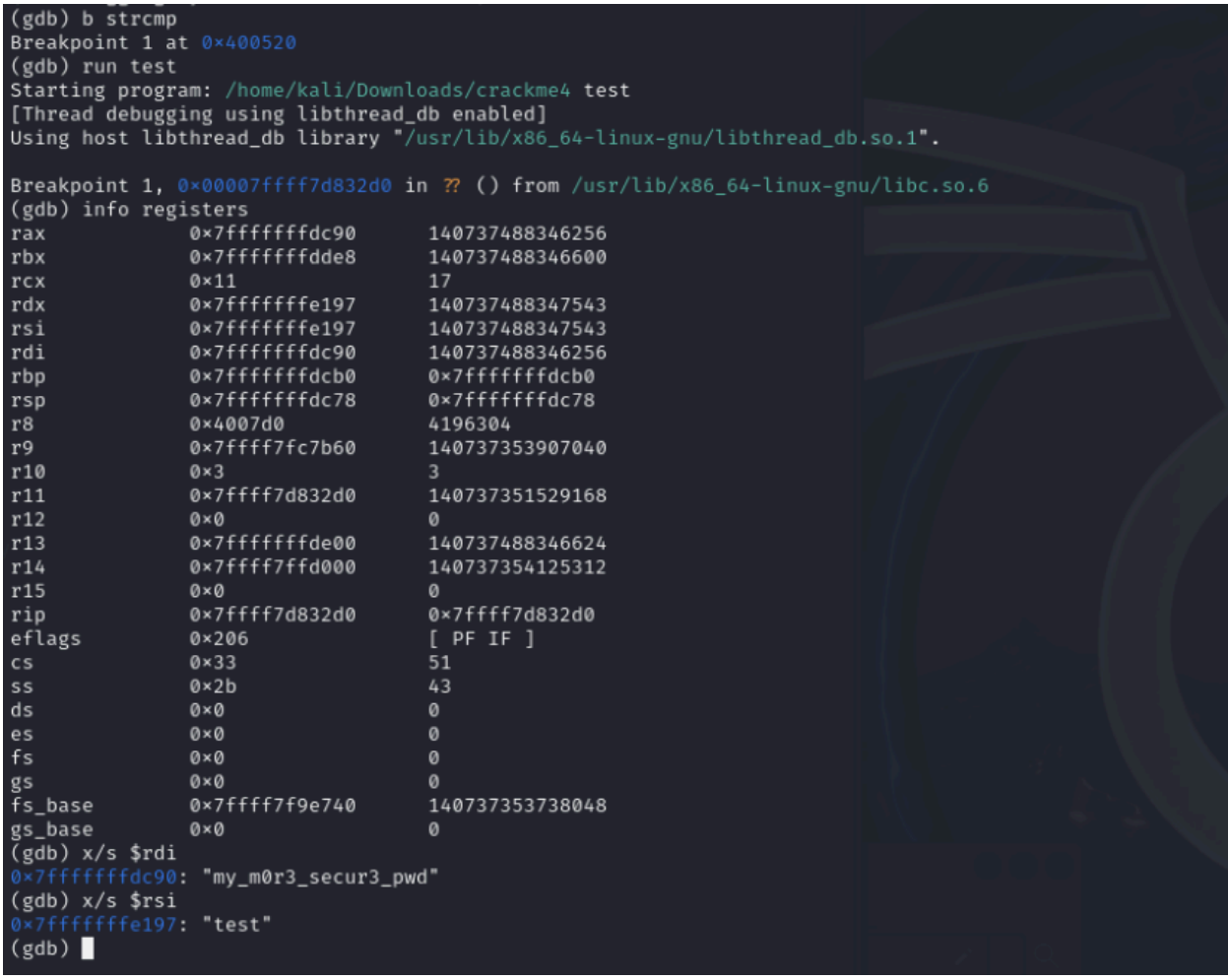
```
/lib64/ld-linux-x86-64.so.2
libc.so.6
puts
__stack_chk_fail
printf
strcmp
__libc_start_main
__gmon_start__
GLIBC_2.4
GLIBC_2.2.5
UH-X
UH-X
I]{I
WAGQV
[]A\A]A^A_
password OK
password "%s" not OK
Usage : %s password
This time the string is hidden and we used strcmp
;*3$"
GCC: (Ubuntu 4.8.2-19ubuntu1) 4.8.2
.symtab
.strtab
.shstrtab
.interp
.note.ABI-tag
.note.gnu.build-id
.gnu.hash
.dynsym
.dynstr
.gnu.version
.gnu.version_r
.rela.dyn
.rela.plt
.init
.text
.fini
.rodata
.eh_frame_hdr
.eh_frame
.init_array
.fini_array
.jcr
.dynamic
.got
.got.plt
.data
.bss
.comment
crtstuff.c
__JCR_LIST__
deregister_tm_clones
register_tm_clones
__do_global_dtors_aux
completed.6972
__do_global_dtors_aux_fini_array_entry
```

4.3.

4.4. Nothing to note

4.5. Try gdb

4.6.



4.6.1. Break at strcmp

4.6.2. You get 400520

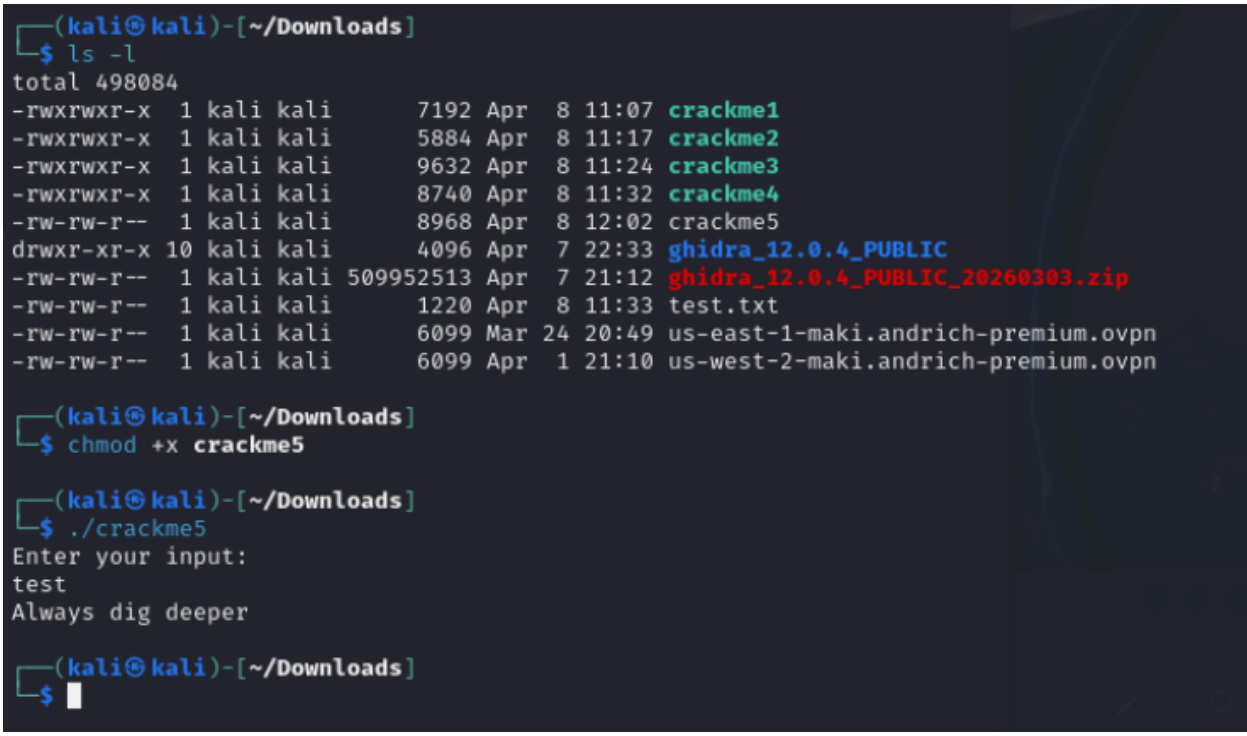
4.6.3. Run test

4.6.4. Then check the values rdi and rsi to see the comparison and you get the password

4.6.5. My_m0r3_secur3_pwd

5. Crackme 5

5.1.



5.1.1. Takes in user input

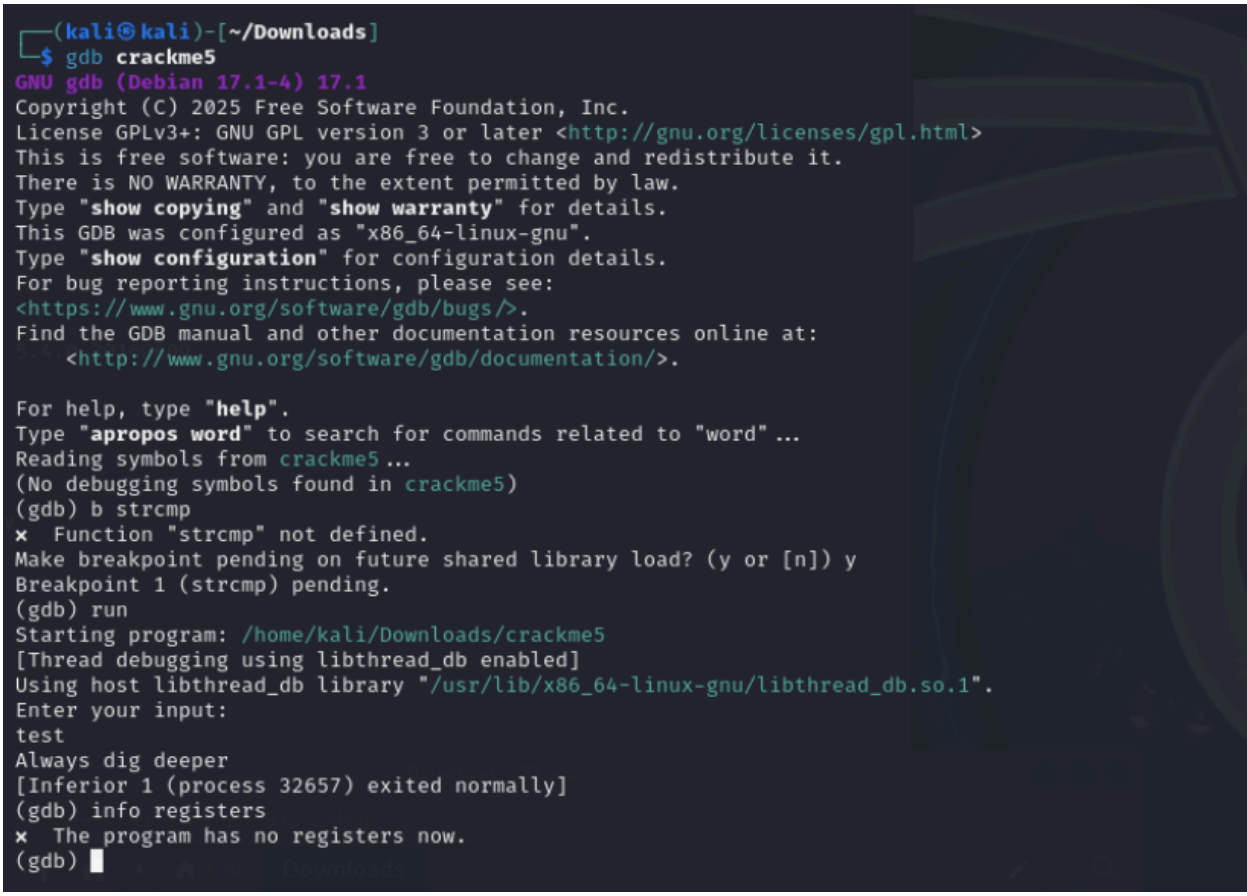
```
/lib64/ld-linux-x86-64.so.2
libc.so.6
strncmp
__isoc99_scanf
puts
__stack_chk_fail
strlen
atoi
__libc_start_main
__gmon_start__
GLIBC_2.7
GLIBC_2.4
GLIBC_2.2.5
UH-`
AWAVA
AUATL
[]A\A]A^A_
Enter your input:
Good game
Always dig deeper
;*3$"
GCC: (Ubuntu 5.4.0-6ubuntu1~16.04.11) 5.4.0 20160609
crtstuff.c
__JCR_LIST__
deregister_tm_clones
__do_global_dtors_aux
completed.7594
__do_global_dtors_aux_fini_array_entry
frame_dummy
__frame_dummy_init_array_entry
strncmp.c
__FRAME_END__
__JCR_END__
__init_array_end
_DYNAMIC
__init_array_start
__GNU_EH_FRAME_HDR
_GLOBAL_OFFSET_TABLE_
__libc_csu_fini
strncmp@@GLIBC_2.2.5
_ITM_deregisterTMCloneTable
puts@@GLIBC_2.2.5
_edata
strlen@@GLIBC_2.2.5
strcmp_
__stack_chk_fail@@GLIBC_2.4
__libc_start_main@@GLIBC_2.2.5
__data_start
__gmon_start__
__dso_handle
_IO_stdin_used
__libc_csu_init
__bss_start
main
_Jv_RegisterClasses
check
```

5.2.

5.2.1. Strings output

5.3. Try gdb

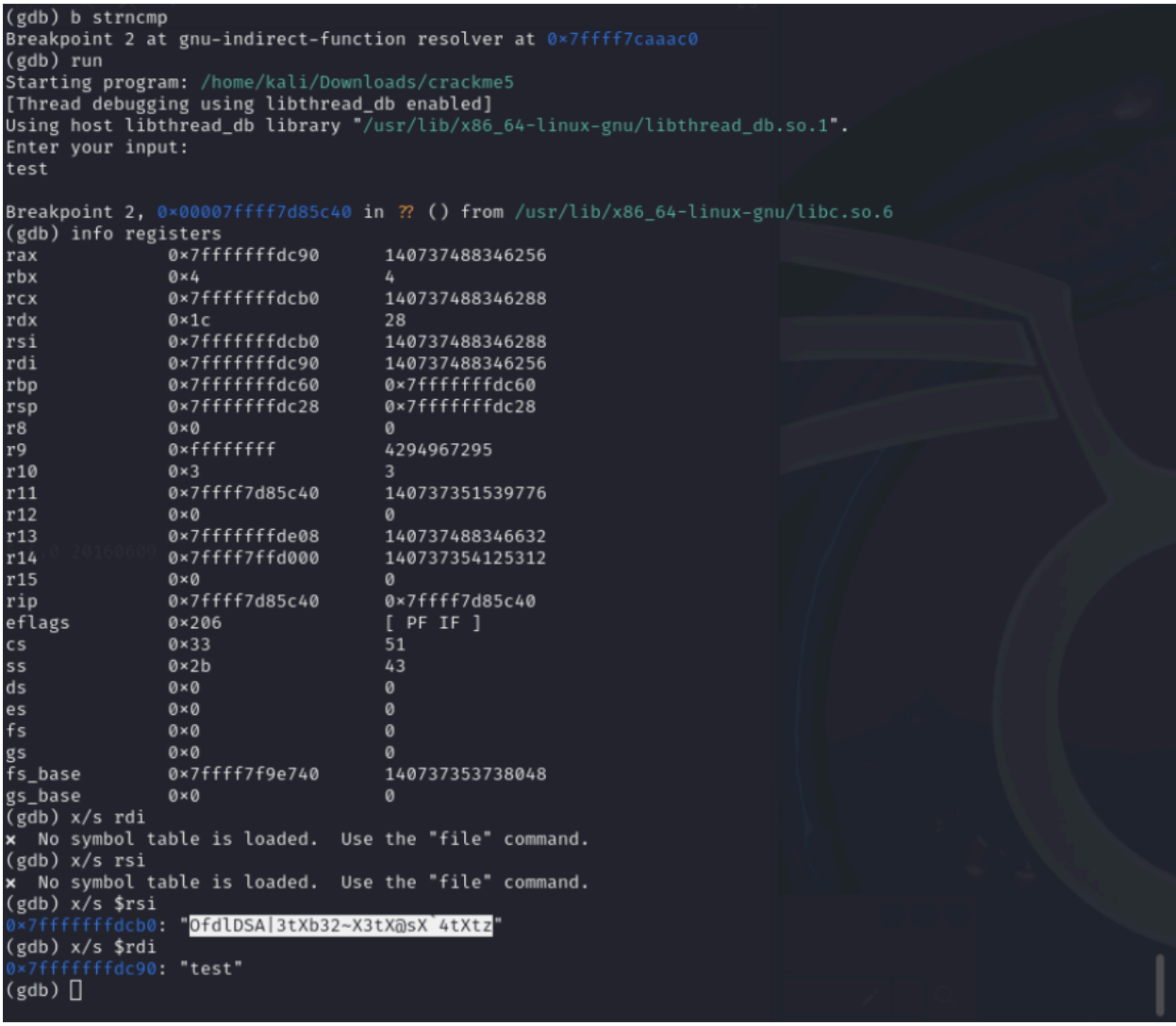
5.4.



5.4.1. Doesnt use strcmp

5.5. Is the strings output you see it uses strncmp

5.6.

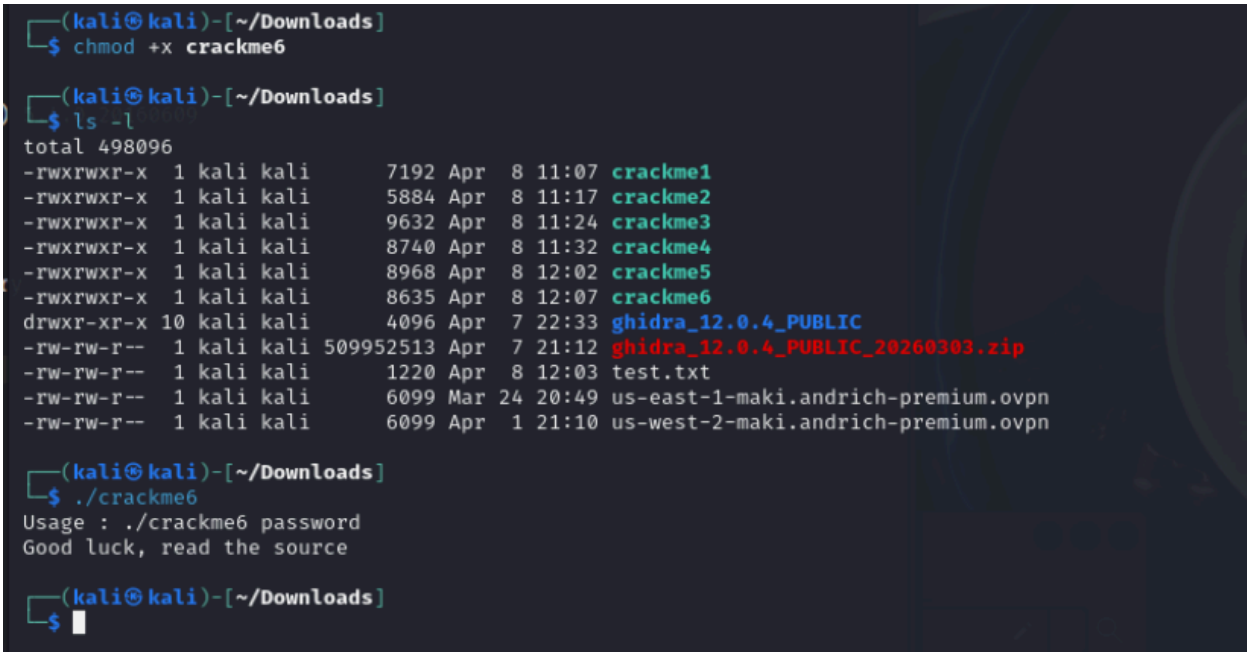


5.6.1. Testings strncmp it worked

5.6.2. And you get the output of OfdlDSA|3tXb32~X3tX@sX`4tXtz

6. Crackme 6

6.1.



6.1.1. Okay we have to read the source

```
 1 /lib64/ld-linux-x86-64.so.2
 2 libc.so.6
 3 puts
 4 printf
 5 __libc_start_main
 6 __gmon_start__
 7 GLIBC_2.2.5
 8 UH-H
 9 UH-H
10 []A\A]A^A_
11 password OK
12 password "%s" not OK
13 Usage : %s password
14 Good luck, read the source
15 ;*3$"
16 GCC: (Ubuntu 4.8.2-19ubuntu1) 4.8.2
17 .symtab
18 .strtab
19 .shstrtab
20 .interp
21 .note.ABI-tag
22 .note.gnu.build-id
23 .gnu.hash
24 .dynsym
25 .dynstr
26 .gnu.version
27 .gnu.version_r
28 .rela.dyn
29 .rela.plt
30 .init
31 .text
32 .fini
33 .rodata
34 .eh_frame_hdr
35 .eh_frame
36 .init_array
37 .fini_array
38 .jcr
39 .dynamic
40 .got
41 .got.plt
42 .data
43 .bss
44 .comment
45 crtstuff.c
46 __JCR_LIST__
47 deregister_tm_clones
48 register_tm_clones
49 __do_global_dtors_aux
50 completed.6972
51 __do_global_dtors_aux_fini_array_entry
52 frame_dummy
```

6.2.1. Nothing useful in the strings

6.3. In ghidra

6.3.1. Find the mysecure test function

6.3.1.1.

```
undefined8 uVar1;

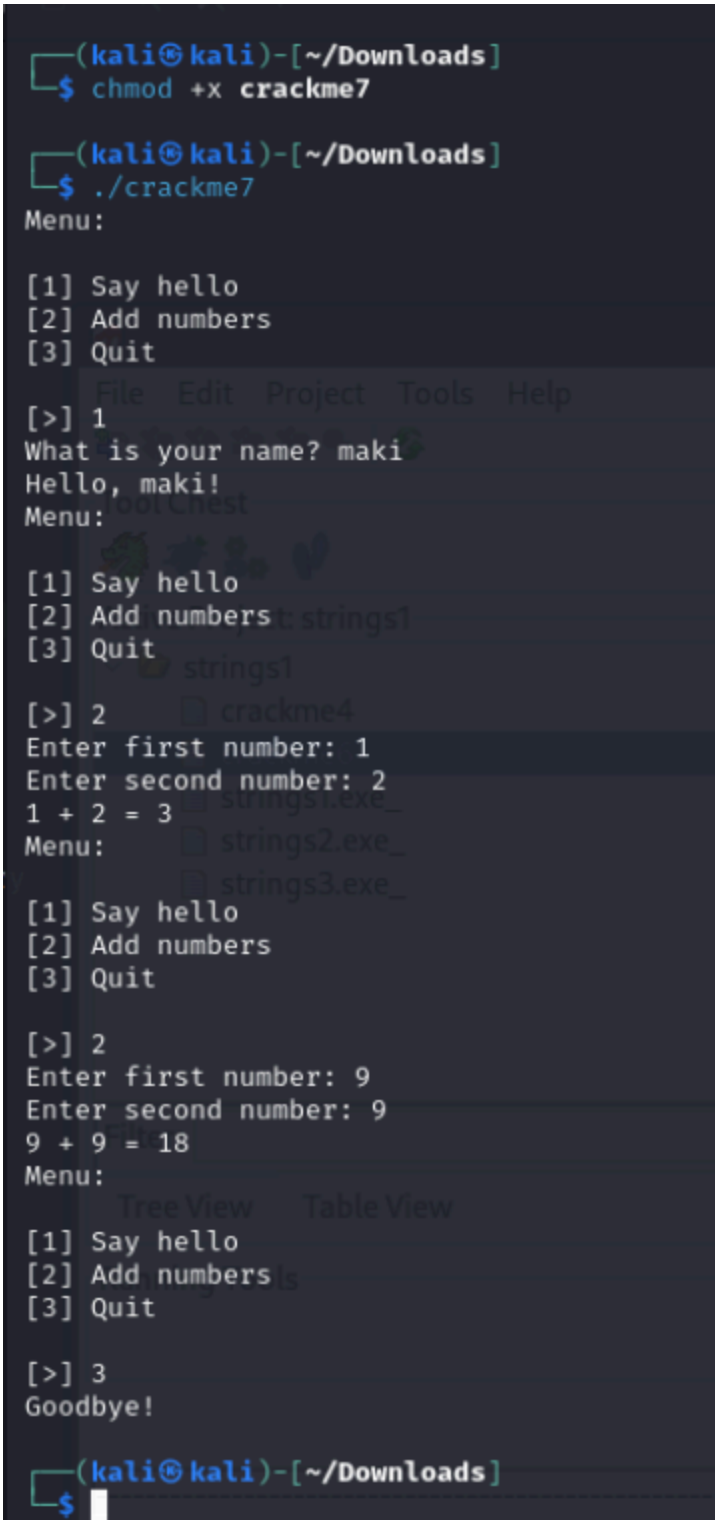
if ((*param_1 == '\0') || (*param_1 != '1')) {
  uVar1 = 0xffffffff;
}
else if ((param_1[1] == '\0') || (param_1[1] != '3')) {
  uVar1 = 0xffffffff;
}
else if ((param_1[2] == '\0') || (param_1[2] != '3')) {
  uVar1 = 0xffffffff;
}
else if ((param_1[3] == '\0') || (param_1[3] != '7')) {
  uVar1 = 0xffffffff;
}
else if ((param_1[4] == '\0') || (param_1[4] != '_')) {
  uVar1 = 0xffffffff;
}
else if ((param_1[5] == '\0') || (param_1[5] != 'p')) {
  uVar1 = 0xffffffff;
}
else if ((param_1[6] == '\0') || (param_1[6] != 'w')) {
  uVar1 = 0xffffffff;
}
else if ((param_1[7] == '\0') || (param_1[7] != 'd')) {
  uVar1 = 0xffffffff;
}
```

6.3.2. You can see the comparison of each character

6.3.2.1. You get 1337_pwd

7. Crackme 7

7.1.



7.1.1. Seems to be a simple math game

7.2.

```
[>] 2
Enter first number: 99999999999999999999999999999999999999999999999
Enter second number: 2
2147483647 + 2 = -2147483647
Menu:

[1] Say hello
[2] Add numbers
[3] Quit

[>] 2
Enter first number: 4534958390453
Enter second number: 423948239423
2147483647 + 2147483647 = -2
```

7.2.1. Will max out at 2147483647

```
tdh
/lib/ld-linux.so.2
libc.so.6
_IO_stdin_used
__isoc99_scanf
puts
printf
memset
__libc_start_main
/usr/local/lib:$ORIGIN
__gmon_start__
GLIBC_2.7
GLIBC_2.0
PTRh
QRPh
j"jA
[^_]
UWVS
t$,U
[^_]
Menu:
[1] Say hello
[2] Add numbers
[3] Quit
[>]
Unknown input!
What is your name?
%99s
Unable to read name!
Hello, %s!
Enter first number:
Unable to read number!
Enter second number:
%d + %d = %d
Goodbye!
Wow such h4x0r!
Unknown choice: %d
;*2$"4
GCC: (Ubuntu 5.4.0-6ubuntu1~16.04.9) 5.4.0 20160609
crtstuff.c
__JCR_LIST__
deregister_tm_clones
__do_global_dtors_aux
completed.7209
__do_global_dtors_aux_fini_array_entry
frame_dummy
__frame_dummy_init_array_entry
loop1.c
giveFlag
__FRAME_END__
__JCR_END__
__init_array_end
```

7.3.1. You see something called give flag….

7.4. Lets Go into gdb and see what we can do

```
(gdb) b main
Breakpoint 2 at 0×80484ca
(gdb) run
Starting program: /home/kali/Downloads/crackme7
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/usr/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 2, 0×080484ca in main ()
(gdb) call giveFlag(0
× A syntax error in expression, near the end of `giveFlag(0'.
(gdb) call giveFlag()
× 'giveFlag' has unknown return type; cast the call to its declared return type
(gdb) call giveFlag
$1 = {<text variable, no debug info>} 0×80486a6 <giveFlag>
(gdb) jump *0×80486a6
Continuing at 0×80486a6.

Breakpoint 1, 0×080486ac in giveFlag ()
(gdb) next
Single stepping until exit from function giveFlag,
which has no line number information.
flag{much_reversing_very_ida_wow}
0×ffffcef0 in ?? ()
(gdb)
```

7.4.1.

7.4.1.1. Set a breakpoint at main

7.4.1.2. When it hits it call the function give flag

7.4.1.3. Confirmed its a func in ghidra

7.4.1.4. It gave us an addy

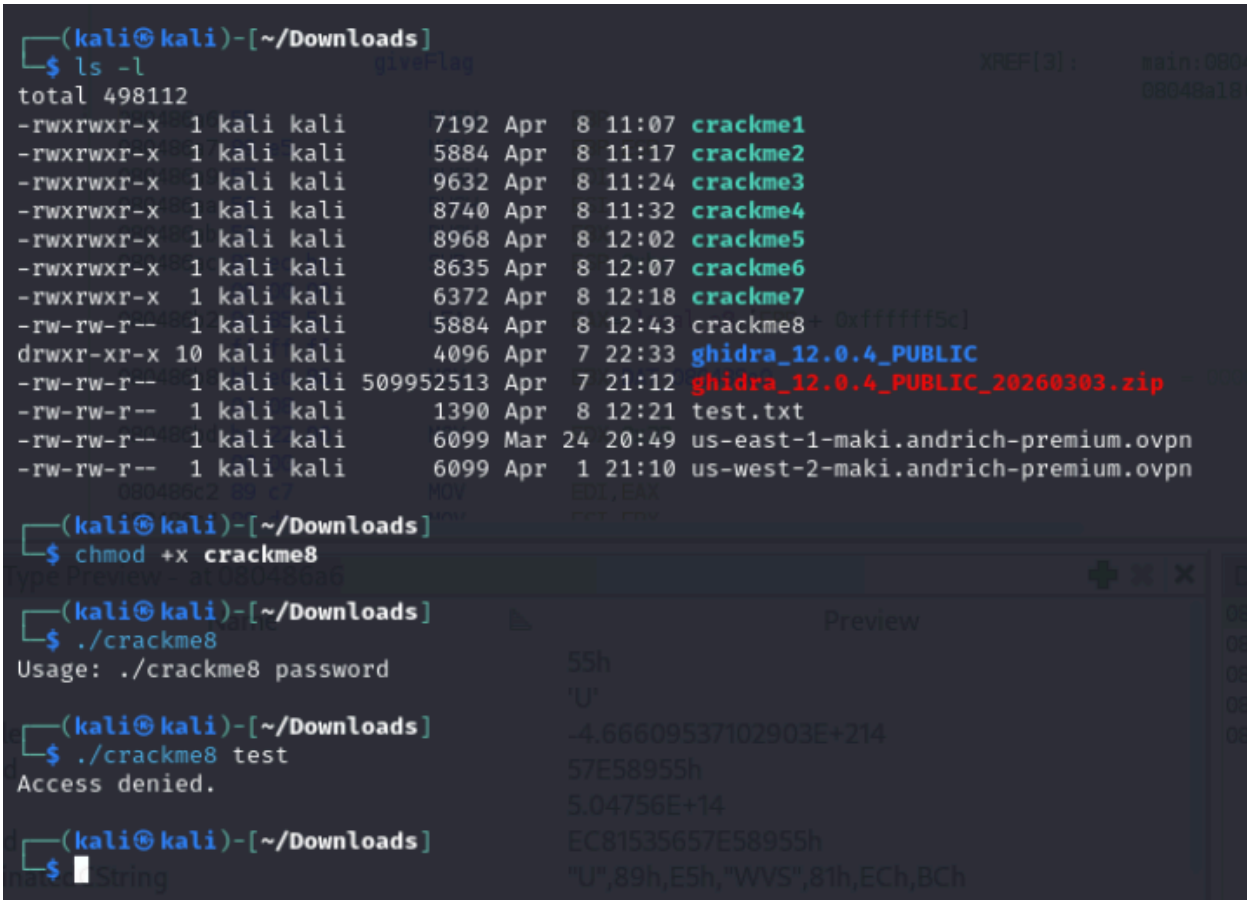
7.4.1.5. Jumped to it

7.4.1.6. Stepped through it and got the flag

7.5. flag{much_reversing_very_ida_wow}

8. Crackme 8

8.1.



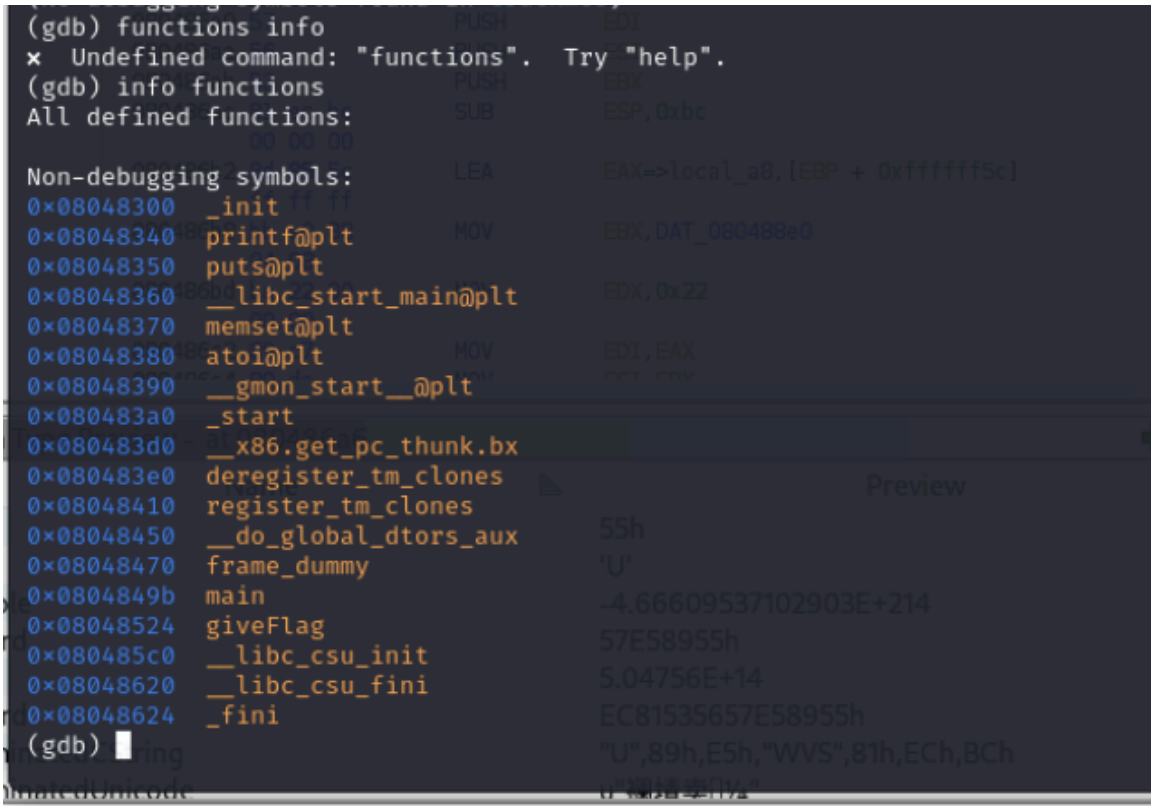
8.1.1. Asks for a password for input

8.1.2.

```
/lib/ld-linux.so.2
libc.so.6
_IO_stdin_used
puts
printf
memset
atoi
__libc_start_main
/usr/local/lib:$ORIGIN
__gmon_start__
GLIBC_2.0
PTRh
j<jA
[^_]
UWVS
t$,U
[^_]
Usage: %s password
Access denied.
Access granted.
;*2$"(
GCC: (Ubuntu 5.4.0-6ubuntu1~16.04.9) 5.4.0 20160609
crtstuff.c
__JCR_LIST__
deregister_tm_clones
__do_global_dtors_aux
completed.7209
__do_global_dtors_aux_fini_array_entry
frame_dummy
__frame_dummy_init_array_entry
conditional2.c
giveFlag
__FRAME_END__
__JCR_END__
__init_array_end
_DYNAMIC
__init_array_start
__GNU_EH_FRAME_HDR
_GLOBAL_OFFSET_TABLE_
__libc_csu_fini
_ITM_deregisterTMCloneTable
__x86.get_pc_thunk.bx
printf@@GLIBC_2.0
_edata
__data_start
puts@@GLIBC_2.0
__gmon_start__
__dso_handle
_IO_stdin_used
__libc_start_main@@GLIBC_2.0
__libc_csu_init
memset@@GLIBC_2.0
_fp_hw
__bss_start
main
_Jv_RegisterClasses
```

8.1.2.1. Strings output

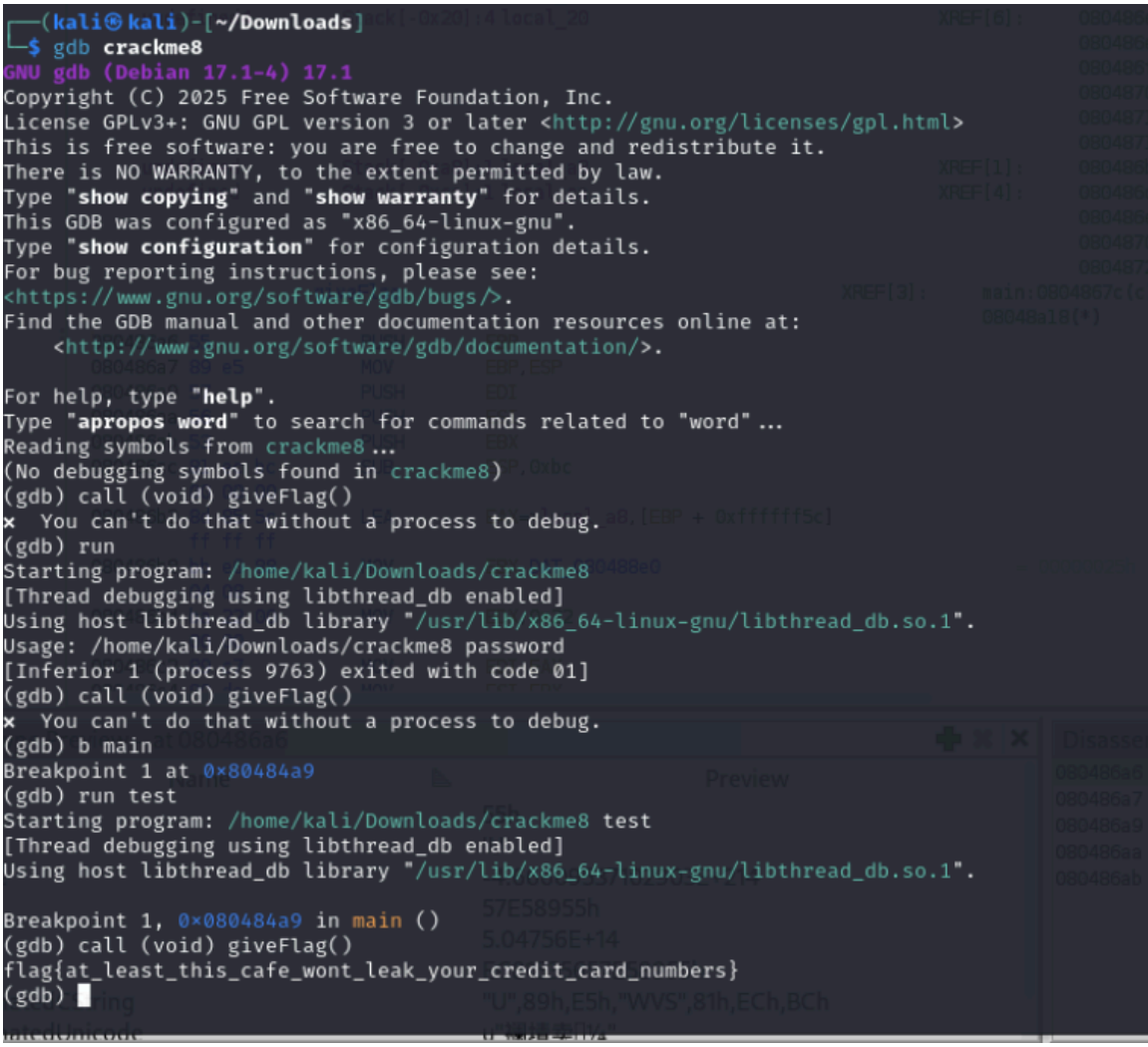
8.2. Check it out in gdb



8.2.1.

8.2.2. Give flag function again

8.2.3.



8.2.3.1. B main

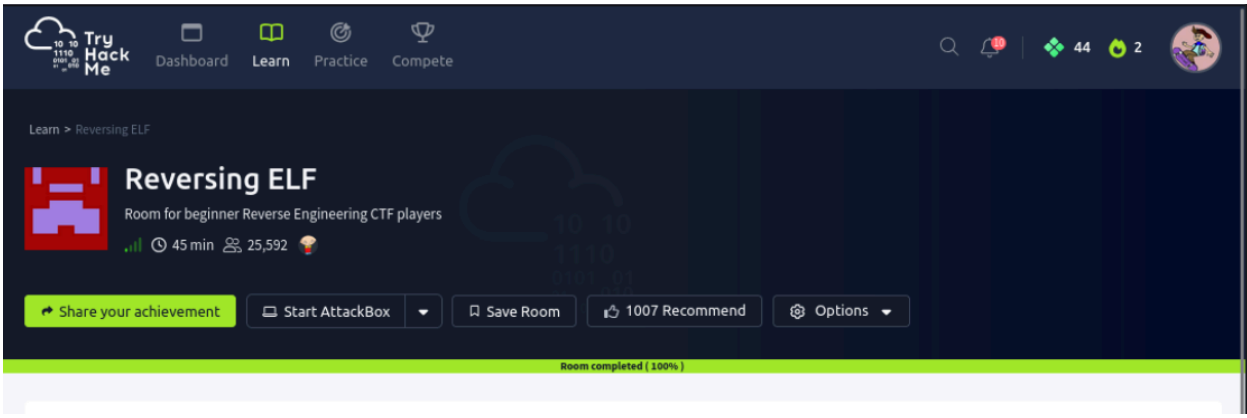
8.2.3.2. Run

8.2.3.3. Call (void) giveFlag

8.2.3.4. Flag{at_least_this_cafe_wont_leak_your_credit_card_numbers}

9. Finish the room

9.1.


Try Hack Me
Dashboard
Learn
Practice
Compete
44
2
Learn > Reversing ELF
Reversing ELF
Room for beginner Reverse Engineering CTF players
45 min
25,592
Share your achievement
Start AttackBox
Save Room
1007 Recommend
Options
Room completed (100%)